

BACKLOG SUBJECT EXAMINATIONS – AUGUST / SEPTEMBER 2023

Program	: B.E :- Computer Science and Engineering	Semester	: III
Course Name	: Data Structures	Max. Marks	: 100
Course Code	: CS33	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT – I

- Illustrate dynamic memory allocation and deallocation with necessary functions. CO1 (06)
 - Write a C program to add two polynomials. CO1 (07)
 - Define Abstract Data Type (ADT) and write an ADT of Natural number. CO1 (07)
- Write a C function to implement recursive binary search. CO1 (04)
 - Analyze sparse matrix representation and write a C function to transpose a sparse matrix. CO1 (08)
 - Explain pattern matching by checking end indices first method with necessary function. CO1 (08)

UNIT – II

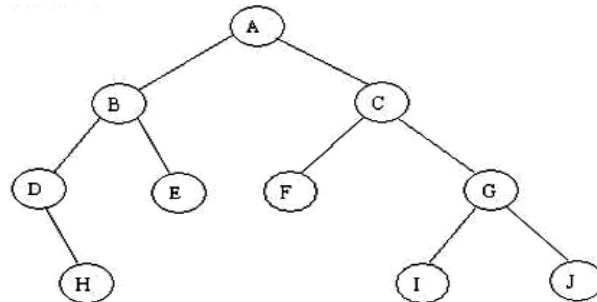
- Write a C function to implement stack using dynamic arrays. CO2 (08)
 - Give the postfix form of the following infix expressions: CO2 (08)
 - $A*B+C$
 - $(A+B)*(C/D)$
 - $A*(B*C+D*E)+F$
 - $(A+B)*C+(D-E)/F+G$
 - Illustrate insert and delete operations on a queue. CO2 (04)
- Write a C function to double the queue capacity of a circular queue. CO2 (06)
 - Explain how to evaluate the following arithmetic expression using stack. CO2 (08)
$$6 \ 2 \ / \ 3 \ - \ 4 \ 2 \ * \ +$$
 - Differentiate between STACK and QUEUE data structures and Write a C function to implement multiple stack. CO2 (06)

UNIT – III

- Write the following C functions to implement singly linked list. CO3 (08)
 - Insert inserting node at the front
 - Delete from a list
 - Printing a list
 - Analyze the advantage of doubly linked list over singly linked list with suitable example. CO3 (08)
 - Illustrate with an example the polynomial representation using list. CO3 (04)
- What is a circular linked list? Write C functions to perform the following operations on circular linked list. CO3 (08)
 - Inserting node at the front
 - Finding length of a list
 - Explain sparse matrix representation with a C code to represent its node structure. CO3 (08)
 - Write a C function to concatenate singly linked lists. CO3 (04)

UNIT – IV

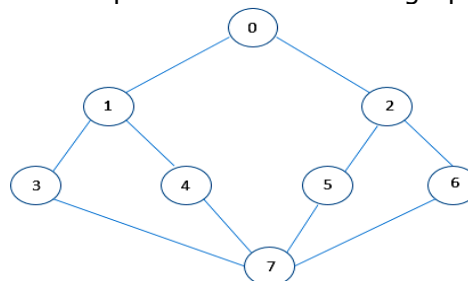
7. a) List and explain the different ways of representing binary tree with a graphical representation. CO4 (06)
- b) Analyze how the binary tree is useful in solving the satisfiability problem. CO4 (07)
- c) How to insert an element into a Binary Search Tree? Write a C function for inserting a dictionary pair into a Binary Search Tree. CO4 (07)
8. a) Traverse the given tree using Inorder, Preorder and Postorder traversals. CO4 (06)



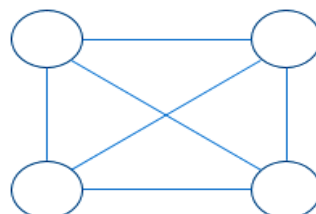
- b) What are threaded binary tree? Write a C function for inserting a node in a threaded binary tree. CO4 (07)
- c) Construct a binary tree from its given inorder and preorder sequences CO4 (07)
 Preorder sequence: ABCDEFGHI
 Inorder sequence: BCAEDGHFI

UNIT- V

9. a) Define Graph and write the ADT of a Graph. CO5 (04)
- b) Give the adjacency list representation of the following graph and write a C function to perform Depth First Search on a graph. CO5 (08)



- c) Name the different types of leftist trees. Illustrate each with an example. CO5 (08)
10. a) What is a spanning Tree? Give the different possible spanning trees of the following graph. CO5 (04)



- b) With a C function illustrate Breadth First Search. CO5 (08)
- c) How does the AVL tree differ from Binary Search Tree? Explain with an example various rotations in AVL trees? CO5 (08)
